

Yui Matsumoto

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TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python

Development Tools: Git, GitHub, Maven, CMake, GNU Make, Visual Studio Code, Neovim

Testing: JUnit 5, JaCoCo

Operating Systems: Linux, Windows, Windows Subsystem for Linux (WSL)

Concepts: Object-Oriented Programming (OOP), UML, Software Testing, Memory Management, Process Management, Multithreading, Synchronization, Socket Programming

TECHNICAL PROJECTS

Dungeon Crawl

Jan 2026 – Apr 2026

Introduction to Software Engineering (CMPT 276), SFU

- Collaborated in a five-member team to develop a 2D dungeon-crawl game in Java using GitHub and Maven.
- Designed UML class diagrams and implemented Java Swing interface components, including popup windows, pause functionality, and outcome-based displays.
- Improved team collaboration by creating project skeleton code, proposing Git branching strategies, and coordinating testing and documentation efforts.
- Contributed to software testing using JUnit 5 and JaCoCo, helping the project achieve 93% instruction coverage and 83% branch coverage.
- Debugged defects, refactored code, and improved project documentation using Javadoc.

Memory Allocator

Jun 2026 – Jul 2026

Systems Programming (CMPT 201), SFU

- Implemented a dynamic memory allocator in C supporting first-fit, best-fit, and worst-fit allocation strategies.
- Designed a custom free-list data structure with block splitting and coalescing of adjacent free blocks.
- Managed heap expansion using `sbrk()` while enforcing configurable heap-size limits.
- Configured project builds using CMake and debugged allocator functionality through testing.

Group Chat Server

Jul 2026 – Aug 2026

Systems Programming (CMPT 201), SFU

- Developed a multi-client group chat server and fuzzing client in C using TCP/IP socket programming.
- Implemented concurrent message transmission and reception while preserving consistent message ordering across connected clients.
- Designed message handling for a binary protocol containing message types, client addresses, ports, and message data.
- Implemented a two-phase termination protocol to coordinate graceful shutdown between the server and multiple clients.

EDUCATION

BSc. in Computing Science

Sep 2025 – Dec 2028

Simon Fraser University, Burnaby, BC